Assignment 5

Due: Friday, November 13, 2026 at 11:59 PM PT

For this assignment, we will use data from:

Lee, K., Miguel, E., & Wolfram, C. (2020). Experimental evidence on the economics of rural electrification. *Journal of Political Economy, 128*(4), 1523-1565. https://doi.org/10.1086/705417

We used these data in Small Assignment 2. We now revisit the setting using randomized controlled trials and the potential-outcomes framework. The variables of interest are:

- TakeUp_i: an indicator equal to 1 if household i connects to the electricity grid and 0 otherwise
- Treatment_i: an indicator equal to 1 if household i is assigned to any subsidy group and 0 if assigned to the control group
- Price_i: the effective connection price after the subsidy
- Earnings_i: monthly earnings of the household head
- HhSize_i: household size
- Female_i: an indicator equal to 1 if the household head is female
- Age_i: age of the household head
- Finschool_i: an indicator equal to 1 if the household head finished secondary school
- Spouse_i: an indicator equal to 1 if the household head is married
- NotFarmer_i: an indicator equal to 1 if the household head is not a farmer
- Employed_i: an indicator equal to 1 if the household head is employed
- Bank_i: an indicator equal to 1 if the household head has a bank account
- OwnLand_i: acres of land owned by the household

All variables except TakeUp_i, Treatment_i, and Price_i were measured at baseline, before the program began.

The randomized price-offer experiment assigned 2,289 initially unconnected households in rural Kenya to four groups:

1. **High subsidy:** 380 households were offered a \$398 (100%) subsidy and faced an effective price of \$0.
2. **Medium subsidy:** 379 households were offered a \$227 (57%) subsidy and faced an effective price of \$171.
3. **Low subsidy:** 380 households were offered a \$114 (29%) subsidy and faced an effective price of \$284.
4. **Control:** 1,150 households received no subsidy and faced the regular connection price of \$398.

The course data file contains 2,504 rows. Of these, 215 have a missing value for `price` and are outside the four price-offer groups listed above. For every analysis in this assignment, restrict the sample to observations with a nonmissing value of `price`; this recovers the 2,289 experimental households.

Question 1

Suppose the researchers had not conducted a randomized controlled trial. Instead, they surveyed households in rural Kenya and compared electricity take-up in places where the government nonrandomly subsidized connections with take-up in places where households faced the regular connection price. Would this comparison identify the causal effect of subsidies on electricity take-up?

Explain why or why not using both words and potential-outcomes notation. How would random assignment change the interpretation of the difference in mean take-up? For this question, group all subsidy levels into one treatment group: $T_i = 1$ if household i is assigned a subsidy and $T_i = 0$ if it is assigned to the control group.

Question 2

a) Before running a regression, show how to estimate the average treatment effect (ATE) of the subsidies, with all subsidy levels grouped together, on the probability of connecting to the grid. Then write the regression that estimates the same quantity.

b) Read `dta_LMW_PS4.dta` using `read_dta()` from the `haven` package. Restrict the data to observations with nonmissing `price`. Compute the ATE as a difference in means, estimate the regression from part (a), and test whether the ATE is statistically significant at the 1% level.

Question 3

a) Check baseline balance between the treatment and control groups. For each of the ten baseline household characteristics (`earn`, `hhsize`, `female`, `age`, `finschool`, `spouse`, `notfarmer`, `employed`, `bank`, and `ownland`), conduct a two-sample t test of the null hypothesis that the group means are equal.

b) Do you conclude that the sample is well balanced? Are any results surprising or concerning? If a baseline variable is imbalanced by chance, how could you address it in the treatment-effect regression?

Question 4

a) What happens when you control for additional pre-treatment covariates in the treatment-effect regression? What are the potential advantages and disadvantages?

b) Re-estimate the treatment-effect regression while controlling for all ten baseline characteristics listed in Question 3. Interpret how the treatment coefficient and its precision change.

Question 5

Households with above-average baseline earnings might respond more strongly to a subsidy because they are better able to pay the remaining connection price. Alternatively, higher-income households might respond less strongly because they need the subsidy less.

Propose, implement, and interpret a test of whether the subsidy effect on take-up differs between households whose head had baseline monthly earnings above versus at or below the sample mean. Write the regression equation, state the null and alternative hypotheses, and interpret the interaction coefficient.

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